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Galaxy groups in the 2dF Galaxy Redshift Survey: luminosity and mass statistics

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ABSTRACT

Several statistics are applied to groups and galaxies in groups in the 2° Field Galaxy Redshift Survey. First, we estimate the luminosity functions for different subsets of galaxies in groups. The results are well fitted by a Schechter function with parameters $M^* - 5\log(h) = -19.90 \pm 0.03$ and $\alpha = -1.13 \pm 0.02$ for all galaxies in groups, which is quite consistent with the results of Norberg et al. for field galaxies. When considering the four different spectral types defined by Madgwick et al. we find that the characteristic magnitude is typically brighter than in the field. We also observe a steeper value, $\alpha = -0.76 \pm 0.03$, of the faint end slope for low star-forming galaxies when compared with the corresponding field value. This steepening is more conspicuous, $\alpha = -1.10 \pm 0.06$, for those galaxies in more massive groups ($\mathcal{M} \gtrsim 10^{14}\, h^{-1}\, \mathrm{M}_\odot$) than that obtained in the lower-mass subset, $\alpha = -0.71 \pm 0.04$ ($\mathcal{M} < 10^{14}\, h^{-1}\, \mathrm{M}_\odot$).

Secondly, we compute group total luminosities using the prescriptions of Moore, Frenk & White. We define a flux-limited group sample using a new statistical tool developed by Rauzy. The resulting group sample is used to determine the group luminosity function and we find a good agreement with previous determinations and semi-analytical models.

Finally, the group mass function for the flux-limited sample is derived. An excellent agreement is obtained when comparing our determination with analytical predictions over two orders of magnitude in mass.

Key words: galaxies: clusters: general – galaxies: luminosity function, mass function – galaxies: statistics.

1 INTRODUCTION

Groups of galaxies constitute one of the most suitable laboratories for the study of properties of intermediate galaxy density environments and their consequences on the process of galaxy formation and evolution. Furthermore, several hints concerning the large-scale structure of the Universe and how structures evolve in the Universe can be drawn from the statistical studies of groups and their properties. Some of them, for instance the luminosities and morphological types of their member galaxies, are sensitive to the processes of mergers and interactions between individual galaxies inside a potential well. Meanwhile, other properties such as group abundance as a function of total luminosity or mass, can provide constraints to hierarchical clustering scenarios and cosmological models. In this paper, we provide a robust statistical study concerning the luminosities of group galaxy members, and also on global group properties such as total luminosities and masses, using one of the largest group catalogues at the present. The accuracy of these determinations allows us a fair comparison with analytical and semi-analytical predictions.

Most studies on the environmental dependence of the galaxy luminosity function have been carried out either in the field or in rich clusters using the largest two-dimensional catalogues and small redshift surveys. Regarding field galaxy luminosity functions, several works have been devoted to its determination over the previous decade (Loveday et al. 1992; Marzke, Huchra & Geller 1994; Lin et al. 1996; Zucca et al. 1997; Ratcliffe et al. 1998). With the advent of large redshift surveys such as the 2° Field Galaxy Redshift Survey (2dFGRS) and the Sloan Digital Sky Survey (SDSS), more reliable statistical results have been obtained. The field luminosity functions determined by Blanton et al. (2001) (SDSS) and Norberg et al. (2001) (2dFGRS) show an excellent agreement, finding a luminosity function accurately described by a Schechter function with parameters $M^*_{b_J} - 5\log(h) \simeq -19.66$ and $\alpha \simeq -1.21$. In particular, the 2dFGRS allowed the determination of the field luminosity function for galaxies of different spectral types (Madgwick et al. 2002), finding a systematic steepening of the faint end slope moving from passive ($\alpha = -0.54$) to active ($\alpha = -1.5$) star-forming galaxies, and also a corresponding faintening of M^*.

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A controversial issue concerning the luminosity function of galaxies in clusters is the steepening at the faint end, compared with the field galaxy luminosity function (López-Cruz et al. 1997; Trentham 1997; Valotto et al. 1997). Recently Goto et al. (2002) computed a composed luminosity function using 204 clusters taken from SDSS (York et al. 2000). They found that the slopes of the luminosity functions (LFs) become flatter towards the redder colour band and have a brighter characteristic magnitude and flatter slopes than the field LF.

The lack of statistical results on smaller overdensities such as groups of galaxies is mainly caused by the fact that two-dimensional identification favours the largest overdensities. Analysing a sample of 66 groups of galaxies identified in redshift space, Muriel, Valotto & Lambas (1998) find a flat faint end for the galaxy luminosity function in groups ($\alpha \simeq -1.0$) compared with the luminosity function in clusters where a large relative number of faint galaxies is present. It is important to remark that most of the luminosity function estimations in groups and clusters of galaxies are computed by subtracting background and foreground contamination caused by the lack of spectroscopic information for galaxy members.

One step further, in order to understand the transition between galaxy and galaxy systems luminosities, is the computation of the luminosity function of galaxy groups. Moore, Frenk & White (1993), reanalysing the groups in the Center for Astrophysics (CfA) redshift survey, developed a method for the estimation of the total luminosity of groups identified in magnitude-limited galaxy surveys. This method allowed them to compute the luminosity function of galaxy systems for a sample of 163 groups with at least three members. Another attempt to determine the luminosity function of virialized systems was made by Marinoni, Hudson & Giuricin (2002). They used the Nearby Optical Galaxy (NOG) sample, which comprises $\sim$7000 galaxies with $cz \leqslant 6000$ km s^{-1} and $B \leqslant 14$, finding a very good agreement with the previous determination of Moore et al. (1993).

On the other hand, the abundance of haloes as a function of mass constitutes a key point in both the determination of a cosmological model and the understanding of the structure collapse. At the present, the more popular models for halo abundance are the analytical model of Press & Schechter (1974) for spherical collapse, the Sheth & Tormen (1999) model for ellipsoidal collapse and the fit obtained by Jenkins et al. (2001) from numerical simulations. Many efforts have been made to determine the mass function of galaxy systems from observations (Bahcall & Cen 1993; Biviano et al. 1993; Girardi et al. 1998). Recently, Girardi & Giuricin (2000) have computed the mass function for a sample of nearby loose groups by Garcia (1993). They found the group mass function to be a smooth extrapolation of the cluster mass function and in reasonable agreement with the predictions of Press & Schechter (1974).

Currently, one of the largest group catalogues was constructed by Merchán & Zandivarez (2002). They have identified groups on the 2dF public 100k data release using a modified Huchra & Geller (1982) group-finding algorithm that takes into account 2dF magnitude limit and redshift completeness masks. This catalogue constitutes a large and suitable sample for both the study of processes in the group environment and the properties of the group population itself. The global effects of the group environment on star formation were analysed by Martínez et al. (2002) using this catalogue. They have found a strong correlation between the relative fraction of different galaxy types and the parent group virial mass. For groups with $M \gtrsim 10^{13}$ M$_\odot$ the relative fraction of star-forming galaxies is significantly suppressed, indicating that even intermediate-mass environments affect star formation. Domínguez et al. (2002) presented hints toward understanding local environment effects affecting the spectral types of galaxies in groups by studying the relative fractions of different spectral types as a function of the projected local galaxy density and the group-centric distance. A similar analysis was performed in known galaxy clusters and their environments in the 2dFGRS by Lewis et al. (2002).

The aim of this work is to use the group catalogue of Merchán & Zandivarez (2002) to obtain reliable determinations of internal and global properties of groups: luminosity functions of galaxies in groups, group luminosity and mass functions. The outline of this paper is as follows. In Section 2, we present a revised version of the 2dF Galaxy Group Catalogue (2dFGGC) used throughout this work. Section 3 describes the methods and results of the luminosity function of galaxies in groups, while in Section 4 the description corresponds to the luminosity function of galaxy groups. The computation of the group mass function and a comparison with analytical models are presented in Section 5. Finally, in Section 6 we summarize our conclusions.

2 THE 2DFGGC

Samples of galaxies and groups used in this work are a new sample constructed using a revised version of the masks and mask software of the 2dFGRS 100k data release, which includes the μ-masks described in Colless et al. (2001) (see http://msowww.anu.edu.au/2dFGRS/Public/Release/Masks/index.html). The group catalogue is obtained following the same procedure as described by Merchán & Zandivarez (2002). In the previous work, the finder algorithm used for group identification is similar to that developed by Huchra & Geller (1982) but modified in order to take into account the redshift completeness and the magnitude limit mask present on the current release of galaxies. Here we include the effect introduced by the magnitude completeness mask (μ-mask). In the construction of the 2dFGGC values of $\delta\rho/\rho = 80$ and $V_0 = 200$ km s^{-1} were used to maximize the group accuracy. The revised group catalogue comprises a total number of 2198 galaxy groups with at least four members and mean radial velocities in the range $900 \leqslant V \leqslant 75\,000$ km s^{-1}. These groups have a mean velocity dispersion of 265 km s^{-1}, a mean virial mass of $9.1 \times 10^{13}\,h^{-1}$ M$_\odot$ and a mean virial radius of $1.15\,h^{-1}$ Mpc. These results show that the new identification keeps the mean properties of the group catalogue obtained by Merchán & Zandivarez (2002), as expected from the fact that the introduction of the magnitude completeness mask is a second-order correction.

3 LUMINOSITY FUNCTION OF GALAXIES IN GROUPS

In this section we estimate the luminosity functions of galaxies in groups. The following analysis comprises the study of the whole sample of galaxies in groups and different subsets defined by galaxy spectral types as defined by Madgwick et al. (2002). This classification is based on the η parameter, which is very tightly correlated with the equivalent width of the H$_\alpha$ emission line, correlates well with morphology for emission-line galaxies and can be interpreted as a measure of the relative current star formation present in each galaxy. The four spectral types are defined as:

(i) type 1: $\eta < -1.4$,
(ii) type 2: $-1.4 \leqslant \eta < 1.1$,
(iii) type 3: $1.1 \leqslant \eta < 3.5$,
(iv) type 4: $\eta \geqslant 3.5$.

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The type 1 class is characterized by an old stellar population and strong absorption features, types 2 and 3 comprise spiral galaxies with increasing star formation, finally the type 4 class is dominated by particularly active galaxies such as starburst.

3.1 Luminosity function estimators

In a comparative study of different LF estimators using Monte Carlo simulations, Willmer (1997) shows that the C^- method of Lynden-Bell (1971) and the STY method derived by Sandage, Tammann & Yahil (1979) are the best estimators to measure the shape of the LF. Moreover, Willmer's analysis states that the C^- is the most robust estimator, being less affected by different values of the faint end slope of the Schechter parametrization and the sample size. In this work we use both the C^- method to make a non-parametric determination of the LF and the STY method to calculate the maximum-likelihood Schechter fit to the LF.

The C^- method was simplified and developed by Choloniewski (1987) in order to compute simultaneously the shape and normalization of the luminosity function. The LF is obtained by differentiating the cumulative LF, $\Psi(M)$. The function $X(M)$ defined as the observed density of galaxies with absolute magnitude brighter than M, represents only an undersampling of the $\Psi(M)$,

$$\frac{\mathrm{d}X}{X} < \frac{\mathrm{d}\Psi}{\Psi}; \tag{1}$$

the key point of the method is to construct a quantity $C(M)$, subsample of $X(M)$, such that

$$\frac{\mathrm{d}X}{C} = \frac{\mathrm{d}\Psi}{\Psi}. \tag{2}$$

Lynden-Bell (1971) defined this quantity as the number of galaxies brighter than M which could have been observed if their magnitude were M. Following Choloniewski (1987), and taking into account the sky coverage of the 2dF present release, the differential LF can be written as

$$\langle\Phi(M)\rangle = \frac{\Gamma \sum_i^{M_i \in [M, M+\Delta M]} \psi_i}{\Delta M}, \tag{3}$$

where

$$\Gamma = \prod_{k=2}^{N} \frac{C_k + w_k}{C_k} \left[V \sum_{i=1}^{N} \psi_i \sum_{j=1}^{N} \frac{R(\alpha_j, \delta_j)}{N} \right]^{-1}, \tag{4}$$

$$\psi_k = \prod_{i=1}^{k} \frac{C_i + w_i}{C_{i+1}} \tag{5}$$

and $R(\alpha, \delta)$ is the redshift completeness of the parent catalogue. $C_k \equiv C^-(M_k)$ is defined as $C(M)$ but excluding the object k itself and weighting each object by the inverse of the magnitude-dependent redshift completeness defined as $w^{-1} = c_z(b_J, \mu) = 0.99[1 - \exp(b_J - \mu)]$ where $\mu = \mu(\alpha, \delta)$ (see Norberg et al. 2001).

In addition to the non-parametric method described above we also apply the STY method that uses a maximum-likelihood technique to find the most probable parameters of an analytical $\Phi(M)$, in general assumed to be a Schechter function:

$$\Phi(M) \propto 10^{-0.4(M-M^*)(\alpha+1)} \exp\left[-10^{-0.4(M-M^*)}\right]. \tag{6}$$

The probability p_i of having an object with absolute magnitude M_i is

$$p_i \equiv \frac{\Phi(M_i)}{\int_{M_{\mathrm{bright}}}^{M_{\mathrm{faint}}} \Phi(M)\,\mathrm{d}M}, \tag{7}$$

where M_{bright} and M_{faint} are the brightest and faintest absolute magnitudes observable in the sample at the redshift of the considered galaxy with the corresponding $k + e$ correction. Consequently, the method seeks parameters M^* and α maximizing the likelihood function

$$\mathcal{L} = \prod_{i=1}^{N} p_i. \tag{8}$$

As we did before, we use the μ-mask to weight each galaxy by $1/c_z(b_J, \mu)$ to compensate for the magnitude-dependent incompleteness. The errors can be estimated from the error ellipsoid defined as

$$\ln \mathcal{L} = \ln \mathcal{L}_{\max} - \frac{1}{2}\chi^2_\beta(N), \tag{9}$$

where $\chi^2_\beta(N)$ is the β point of the χ^2 distribution with N degrees of freedom.

The comoving volumes involved in previous equations are estimated using the general formula

$$V = \frac{\omega c}{H_0} \int_{z_1}^{z_2} \frac{d_L(z)^2 (1+z)^{-2}\,\mathrm{d}z}{\sqrt{\Omega_0(1+z)^3 + (1 - \Omega_0 - \Omega_\Lambda)(1+z)^2 + \Omega_\Lambda}}, \tag{10}$$

where ω is the solid angle, c is the speed of light, $H_0 = 100\,h$ km s^{-1} Mpc^{-1} and $d_L(z)$ is the luminosity distance. Hereafter we adopt the cosmological model $\Omega_0 = 0.3$ and $\Omega_\Lambda = 0.7$.

3.2 Luminosity function results

In Fig. 1 we show the luminosity function for all galaxies in groups in arbitrary units. It should be taken into account that the normalization procedure is only necessary for the group LF. Absolute magnitudes are computed using the $k + e$ mean correction for galaxies in the

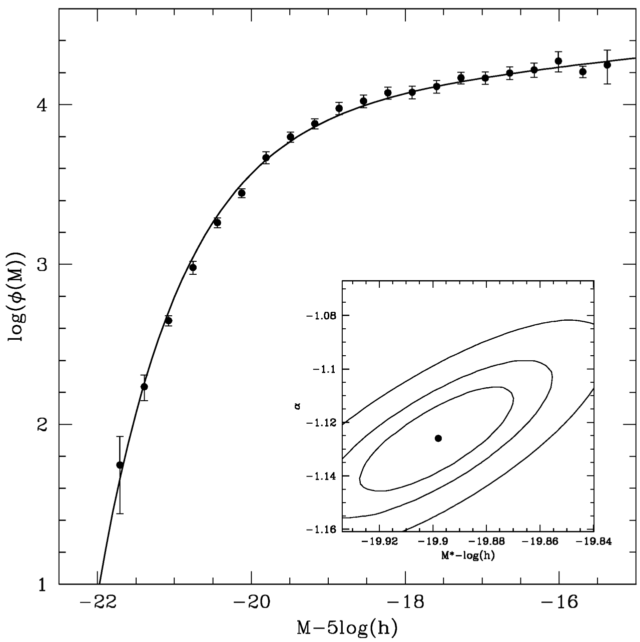

Figure 1. b_J-band luminosity function of galaxies in groups from the 2dFGGC (arbitrary units) computed using the C^- method. Error bars were estimated using mock catalogues as described in the text. The solid line shows the STY Schechter fit to the data. The inset panel displays 1σ, 2σ and 3σ confidence ellipses enclosing the best-fitting Schechter function parameters $\alpha = -1.13 \pm 0.02$ and $M^* - 5\log h = -19.90 \pm 0.03$.

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2dFGRS as derived by Norberg et al. (2001). Error bars are estimated using 10 mock catalogues constructed from numerical simulations of a cold dark matter universe according to the cosmological model adopted in this work with a Hubble constant $h = 0.7$ and a relative mass fluctuation $\sigma_8 = 0.9$. These simulations were performed using 128^3 particles in a cubic comoving volume of $180\,h^{-1}$ Mpc per side. The solid line shows the STY best fit which corresponds to the Schechter parameters $M^* = -19.90 \pm 0.03$ and $\alpha = -1.13 \pm 0.02$. The quoted errors are the projections of the 1σ joint error ellipse (inset plot) on to each axis. The inset plot of Fig. 1 also shows the error ellipsoids defined by 2σ and 3σ error dispersion.

As pointed out by Martínez et al. (2002), the relative fraction of the spectral types in groups and fields are different, with type 1 galaxies being the dominant population in groups. We are interested in deepening this result by studying the LF for the four spectral types in groups. The luminosity functions for different galaxy spectral types are shown in Fig. 2. In these computations we applied a redshift cut-off of $z < 0.15$ since η determinations are available only for galaxies in this redshift range. In this case, error bars were estimated using the bootstrap resampling technique. The corresponding STY best-fitting parameters are quoted in Table 1, and were computed using only those galaxies brighter than $M_{b_J} - 5\log h = -17$, avoiding points with larger error bars.

Table 1. STY estimates of the Schechter parameters for the luminosity functions of galaxies in groups.

Sample	Redshift range	Number of galaxies	$M^* - 5\log(h)$	α
All	0.003–0.25	14 620	-19.90 ± 0.03	-1.13 ± 0.02
Type 1	0.003–0.15	6 202	-19.85 ± 0.04	-0.76 ± 0.03
Type 2	0.003–0.15	3 445	-19.54 ± 0.05	-0.94 ± 0.04
Type 3	0.003–0.15	1 871	-19.37 ± 0.08	-1.25 ± 0.05
Type 4	0.003–0.15	952	-19.40 ± 0.10	-1.40 ± 0.07

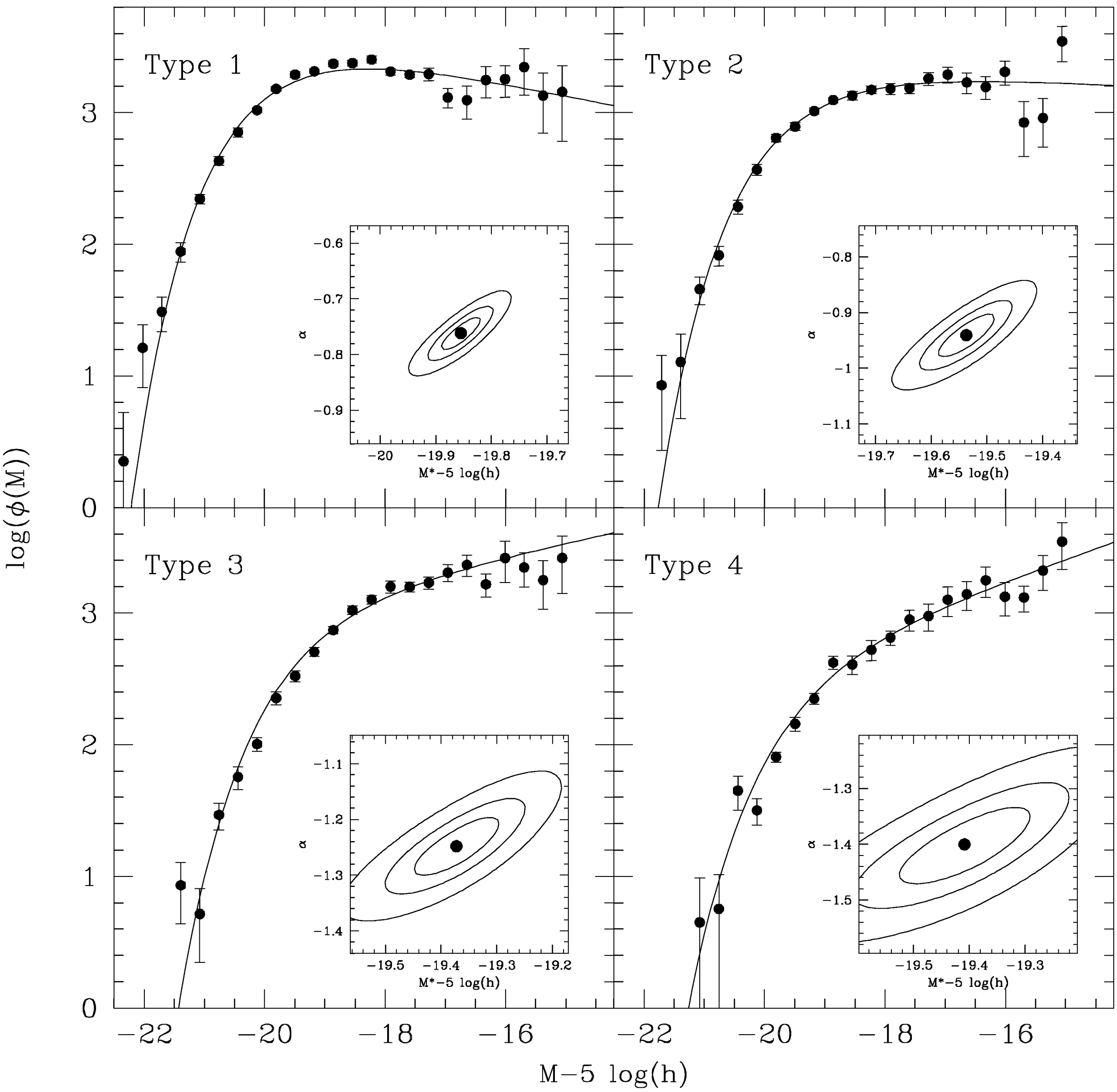

Figure 2. b_J-band luminosity function per spectral type for galaxies in groups from the 2dFGGC (arbitrary units) computed using the C^- method. Error bars are bootstrap resampling technique estimates. Solid lines show the STY Schechter fits to the data. The inset panels display 1σ, 2σ and 3σ confidence ellipses enclosing the best-fitting Schechter function parameters, see Table 1.

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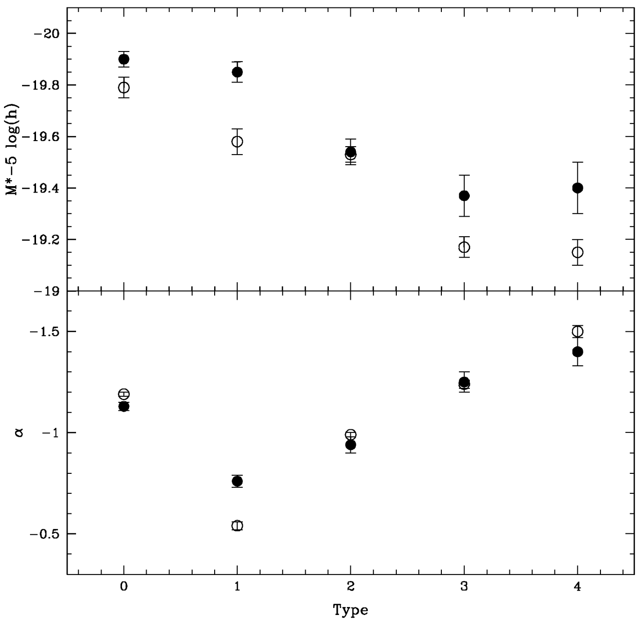

Figure 3. Comparison between the LF Schechter parameters of field galaxies by Madgwick et al. (2002) (open circles) and galaxies in groups of this work (filled circles). The upper panel compares the characteristic magnitude $M^* - 5\log(h)$, meanwhile the lower panel is the comparison for the faint-end slope α. Both panels are plotted as a function of galaxy spectral type. We label as type 0 all galaxies, irrespective of spectral type.

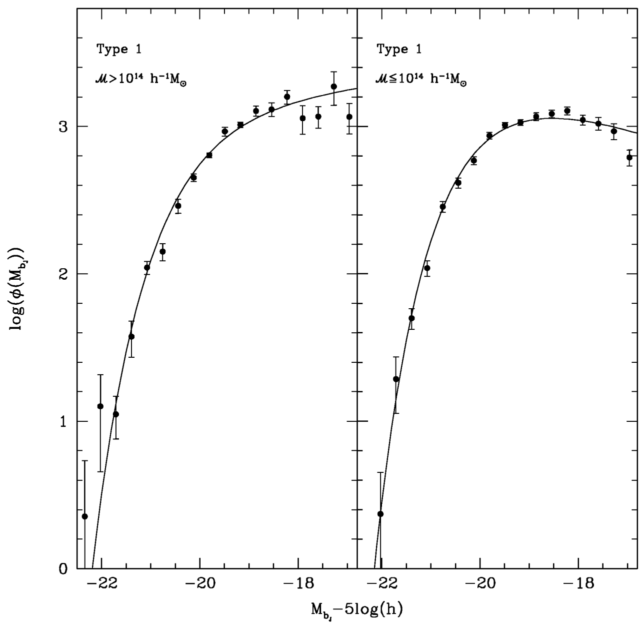

Figure 4. Luminosity functions for type 1 galaxies in high- (left-hand panel) and low- (right-hand panel) mass group samples. Solid lines are the best Schechter fit to the luminosity function determined with the STY method.

At this point, a straightforward comparison between our results and those obtained for field galaxies in the 2dFGRS by Madgwick et al. (2002) can be made. The comparison between the LF Schechter parameters is shown in Fig. 3. Overall, a brightening of the characteristic magnitude is observed for galaxies in groups with a statistical significance of $\sim 2\sigma$. This is particularly significant for type 1 galaxies, which are the main contributors to the general LF of galaxies in groups (3σ). The denser environment of groups could be thought of as the natural cause of this brightening. In this particular environment, galaxy mergers seems to be the most probable cause since tidal interactions are more effective when the galaxy encounter is slow. The latter situation is more frequently observed in groups where smaller velocity dispersions contrast with the high-velocity dispersions observed in rich clusters. Other processes, such as ram-pressure (Abadi, Moore & Bower 1999) or galaxy harassment (Moore et al. 1996) should not be as effective as in rich clusters where the intracluster gas is much denser and the interaction rate is higher.

Nevertheless, the processes mentioned above may be important in the generation of red low-mass galaxies, which are possibly the remnants of dynamical stripped galaxies in high-mass systems and can be the responsible for the steepening of the luminosity function in clusters. The observed difference in the α parameter for type 1 galaxies (lower panel of Fig. 3) may be explained within this framework. In order to test this scenario, we reanalyse the LF for non-star-forming galaxies (type 1) splitting the sample into high ($M \geqslant 10^{14}\,\mathrm{M}_\odot$) and low ($M < 10^{14}\,\mathrm{M}_\odot$) group mass subsets. A significant difference is observed at the faint end slope of the LF from this comparison, resulting in Schechter α parameters of (-1.10 ± 0.06) and (-0.71 ± 0.04) for high- and low-mass subsamples, respectively (Fig. 4). These results may support our hypothesis that in groups, mergers are probably responsible for the M^* brightening, meanwhile processes such as the ram-pressure and galaxy harassment could generate the increase at the LF faint end slope as observed for non-star-forming galaxies in high-mass systems. Besides the steepening of the faint end slope of type 1 galaxies LF in higher-mass systems, the galaxy LF for high- and low-mass groups do not differ appreciably between them, showing roughly the same behaviour as the overall LF of galaxies in groups (Fig. 1).

Despite the differences between the resulting luminosity functions per spectral type, the general LF of galaxies in groups is quite similar to that of 2dF field galaxies obtained by Norberg et al. (2001) and Madgwick et al. (2002). This behaviour could be possibly caused by the different relative abundances of type 1 galaxies in the field and groups. In groups type 1 galaxies are by far the dominant population, consequently, these galaxies almost determine M^* by themselves. Meanwhile, in the field, types 2–4 contribute more strongly to the LF, generating an M^* brighter than that corresponding to type 1 galaxies alone. Regarding the faint end slope of the overall LF, the main contributors are late-type galaxies (except for high-mass groups, as stated above) in both the field and groups. Since there are no significant differences between the α parameters for late types in the field and groups, then the overall faint end slopes are similar.

4 LUMINOSITY FUNCTION OF GROUPS

The next step in this work is to analyse the luminosity distribution of groups independently of the detailed arrangement of luminous material within each object. In this section we determine individual group luminosities that allow us to compute the group luminosity function and the subsequent comparison with previous determinations and predictions from semi-analytical models of galaxy formation.

4.1 Group luminosities

We compute group luminosities following the prescription by Moore et al. (1993). For each group the total luminosity, L_{tot}, is the sum of

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the luminosities of the galaxy members ($L_{\rm obs}$) plus the integrated luminosity of galaxies below the magnitude limit of the survey ($L_{\rm cor}$),

$$L_{\rm tot} = L_{\rm obs} + L_{\rm cor}. \tag{11}$$

In the computation of $L_{\rm cor}$ we assume that group members are independently drawn from the previously computed Schechter fit to the luminosity function of galaxies in groups. Consequently, the expected luminosity of faint group members is

$$L_{\rm cor} = N_{\rm obs} \frac{\int_0^{L_{\rm lim}} L\Phi(L)\,{\rm d}L}{\int_{L_{\rm lim}}^{\infty} \Phi(L)\,{\rm d}L}, \tag{12}$$

where $L_{\rm lim} = 10^{0.4(\mathrm{M}_\odot - M_{\rm lim})}$, $M_{\rm lim} = m_{\rm lim} - 25 - 5\log[d_L(z)]$ and $\mathrm{M}_\odot = 5.30$ (b_J band). The reliability of this scheme for computing luminosities of virialized systems was widely tested by Moore et al. (1993) using groups from the CfA catalogue and flux- and volume-limited mock catalogues. In order to apply this procedure to our group catalogue we introduce some changes that take into account the particular sky coverage of the parent catalogue: $m_{\rm lim}$ is a function of the angular position of the group and $N_{\rm obs} = \sum_{i=1}^{N} w_i$, where the sum extends over the group members and each member is weighted with the inverse of the redshift completeness, $w_i = 1/R(\alpha_i, \delta_i)$, at its angular position.

4.2 Group luminosity function

As a first step in computing the group luminosity function using the C^- method, it is necessary to find the completeness limit in group apparent magnitude of our catalogue. Rauzy (2001) proposed a new tool in order to define this completeness limit for a magnitude–redshift sample. This method does not presuppose that the galaxy population does not evolve with time and is homogeneously distributed in space as for the $V/V_{\rm max}$ test of Schmidt (1968). Rauzy's method assumes that the sample is complete in apparent magnitude up to a given magnitude limit $m_{\rm lim}$. The limiting magnitude determines a correlation between the absolute magnitude M and the $(k+e)$-corrected distance modulus Z. This method is based on the definition of a random variable

$$\zeta \equiv \frac{\Psi(M)}{\Psi[M_{\rm lim}(Z)]}, \tag{13}$$

where $M_{\rm lim}(Z) = m_{\rm lim} - Z$. ζ should be uniformly distributed between 0 and 1 and should be statistically independent of Z and the angular selection function. For each object, an unbiased estimate of ζ is provided by

$$\hat{\zeta}_i = \frac{r_i}{n_i + 1}, \tag{14}$$

where r_i is the number of objects in the sample with $M \leqslant M_i$ and $Z \leqslant Z_i$, and n_i is the number of objects such that $M \leqslant M_{\rm lim}^i(Z_i)$ and $Z \leqslant Z_i$. The expectation E_i and variance V_i of the $\hat{\zeta}_i$ are $\frac{1}{2}$ and $(n_i - 1)/[12(n_i + 1)]$, respectively. Then, the statistic $T_{\rm C}$, defined as

$$T_{\rm C} = \frac{\sum_{i=1}^{N_{\rm gal}} \left(\hat{\zeta}_i - \frac{1}{2}\right)}{\left(\sum_{i=1}^{N_{\rm gal}} V_i\right)^{1/2}} \tag{15}$$

has a mean of zero and unity variance. The test consists in computing the quantity $T_{\rm C}$ on truncated subsamples according to an increasing apparent magnitude limit m_*. While m_* remains below $m_{\rm lim}$ the subsample is complete and $T_{\rm C}$ statistics are distributed with a mean close to zero and unity fluctuations. When m_* becomes greater than $m_{\rm lim}$, the incompleteness introduces a lack of objects with M fainter than $M_{\rm lim}(Z)$, therefore $T_{\rm C}$ is expected to fall systematically to negative values. The behaviour of $T_{\rm C}$ for our group catalogue is shown in the upper panel of Fig. 5, where it can be seen that $T_{\rm C}$ decreases monotonically for limiting apparent magnitudes greater than $m_{\rm lim} \sim 15.6$ (vertical solid line), taking values under $-2(\equiv -2\sigma_{T_{\rm C}})$ (horizontal long dashed lines). The lower panel of the same figure shows the $(k+e)$-corrected distance modulus Z as a function of the absolute magnitude for all groups, the vertical solid line corresponds to the 15.6 limiting apparent magnitude cut-off as determined with the $T_{\rm C}$ analysis. Dashed horizontal line in the lower panel of Fig. 5 represents a low-redshift cut-off ($z_{\rm co} = 0.04$) that we impose in order to avoid small-volume undersampling effects.

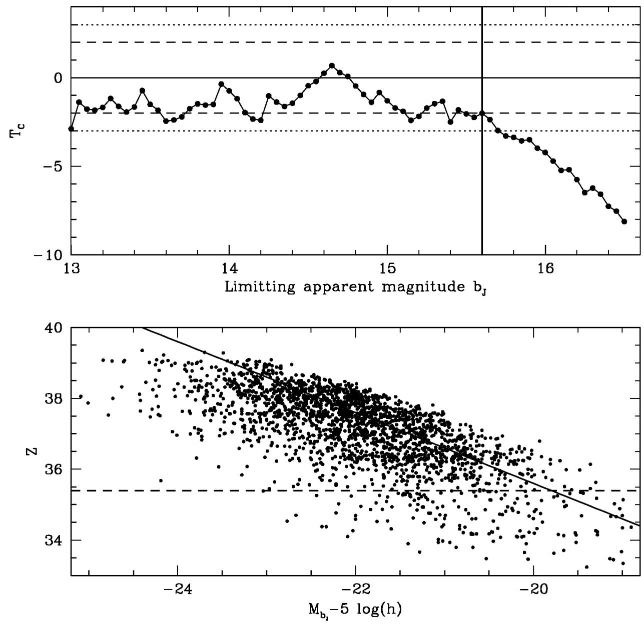

Figure 5. Upper panel, the test for completeness $T_{\rm C}$ (Rauzy 2001) applied to 2dFGGC. Apparent b_J magnitudes were computed using group total luminosities following Moore et al. (1993) and group redshifts. A systematic decline of the $T_{\rm C}$ statistics can be observed beyond $m_{\rm lim} = 15.6$ (vertical line). Lower panel, the distance modulus Z versus the absolute magnitude for groups. The solid line shows the apparent magnitude of completeness $b_J = 15.6$. The horizontal dashed line represents the low-redshift cut-off $z > 0.04$.

Finally, our flux-limited group sample comprises 922 groups with the constraints $0.04 \leqslant z \leqslant 0.25$ and $m_{b_J} \leqslant 15.6$. We compute for this sample the group LF with the C^- method in a similar way as we did for galaxies in groups in Section 3, but in this case no weight is needed in the $C_k \equiv C^-(M_k)$ computation (i.e. $w \equiv 1$ in equations 4 and 5) since the choice of the magnitude cut-off ensures the completeness of the final sample. The resulting group LF is shown in Fig. 6. Error bars were computed by applying the same LF estimator to the mock catalogues in a similar way as explained is Section 3. In the same figure we show the results from groups in the CfA redshift survey by Moore et al. (1993) and from semi-analytical models by Benson et al. (2000) for a Λ cold dark matter model. A general good agreement with previous results is observed except for $M_{b_J} - 5\log h > -21.5$. Since the 2dFGGC has only groups with at least four members, there is a lack of low-luminosity systems that could explain the differences in the low-luminosity tail in Fig. 6 when comparing with previous works. As shown by Moore et al. (1993), the faint end of the group luminosity function is made up almost entirely of single galaxies; binaries and groups with three members link this to the steep bright tail of richer groups and

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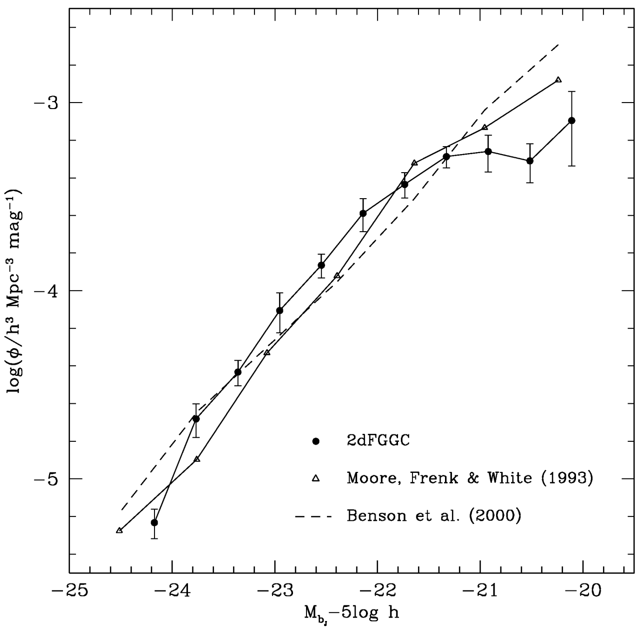

Figure 6. Group luminosity function (solid circles) for our flux-limited group sample compared with the CfA redshifts survey groups (open triangles) Moore et al. (1993) and the semi-analytical models of Benson et al. (2000) (dashed lines).

clusters. We have checked whether the four-member criterion is affecting the faint end of the group luminosity function by recomputing it including groups with three and two members. The resulting group LF is in agreement with the results of Moore et al. (1993) over the whole absolute magnitude range. Nevertheless, it should be recalled that the four-member cut-off is necessary in order to avoid spurious detections. As stated by Merchán & Zandivarez (2002), these false detections can reach approximately ~ 40 per cent when considering groups with only three members. Consequently, the four-member criterion ensures a reliable determination of the luminosity function of virialized systems over a wide range of luminosities.

5 MASS FUNCTION OF GROUPS

The statistical strength of our group sample allows us an accurate measurement of the group mass function. In this section we explore the mass function of intermediate-mass systems, using the flux-limited sample defined in the previous section. For this purpose we use the $1/V_{\max}$ procedure, which is a non-parametric method consisting in weighting each object with the inverse of the maximum comoving volume of the survey, $V_{\max}$,

$$V_{\max}(M_i) = \int_{\min(z_2, z_{\max})}^{\max(z_1, z_{\min})} \frac{\mathrm{d}V}{\mathrm{d}z}\,\mathrm{d}z \tag{16}$$

in which it remains observable given all the observational selection limits. The extremes of the volume integral are found by solving the equations

$$M_i = \begin{cases} m_2 - 5\log d_L(z_{\max}) - 25 - k(z_{\max}) \\ m_1 - 5\log d_L(z_{\min}) - 25 - k(z_{\min}), \end{cases} \tag{17}$$

which are the redshifts of objects with the same absolute magnitude, M_i, of the considered object, but with apparent magnitude at the bright and faint limits of the survey. Consequently, the differential mass function can be computed as

$$\frac{\mathrm{d}N}{\mathrm{d}\mathcal{M}}(\mathcal{M}) = \sum_{|\mathcal{M}_i - \mathcal{M}| \leqslant d\mathcal{M}/2} \frac{1}{V_{\max}(M_i)\hat{R}(\alpha_i, \delta_i)}, \tag{18}$$

where $\mathcal{M}_i$ are the group masses and $\hat{R}(\alpha_i, \delta_i)$ are the mean group redshift completeness.

For a meaningful comparison between the resulting group mass function and analytical mass function predictions (Press & Schechter 1974; Sheth & Tormen 1999; Jenkins et al. 2001), it is necessary to relate the group virial masses with the mass definition in the models. Group virial masses were computed inside a galaxy density contrast $(\delta\rho/\rho)_{\mathrm{g}} = 80$ (Merchán & Zandivarez 2002), which corresponds to a mass density contrast $(\delta\rho/\rho)_m = (\delta\rho/\rho)_{\mathrm{g}}/b$, where b is the linear bias factor. If we estimate $b = 1/\sigma_8$, assuming $\sigma_8 = 0.9$, the resulting density contrast is $(\delta\rho/\rho)_m = 72$. In order to make a comparison with the analytical predictions, the masses should be computed for a particular density contrast depending on the cosmological model. Taking into account the relation between the virial density of collapsed objects in units of the critical density and the adimensional density parameter Ω_0, as quoted by Eke, Cole & Frenk (1996), we must have $(\delta\rho/\rho)_m \sim 330$ for our cosmological model, assuming the spherical collapse approximation. To find the relation between the group virial masses and that corresponding to the matter overdensity 330, we assume a simple mean overdensity profile for groups that scales as r^{-2}, where r is the group-centric distance. Given that in this case $\mathcal{M} \propto r$ then the resulting scaling relation is $\mathcal{M}_{330} \simeq \mathcal{M}_{72}\sqrt{72/330}$.

Fig. 7 shows the comparison between the group mass function with masses scaled using the previous relation, and the analytical prediction for Press & Schechter (1974), Sheth & Tormen (1999) and Jenkins et al. (2001). Overall, analytical predictions are in excellent agreement with our results spanning two orders of magnitude in mass ($10^{13} \lesssim \mathcal{M} \lesssim 10^{15}\,h^{-1}\,\mathrm{M}_{\odot}$). The differences for masses smaller than $\sim 10^{13}\,h^{-1}\,\mathrm{M}_{\odot}$ arise as a consequence of the lack of low-mass groups, mainly caused by the member number cut-off imposed on the 2dFGGC.

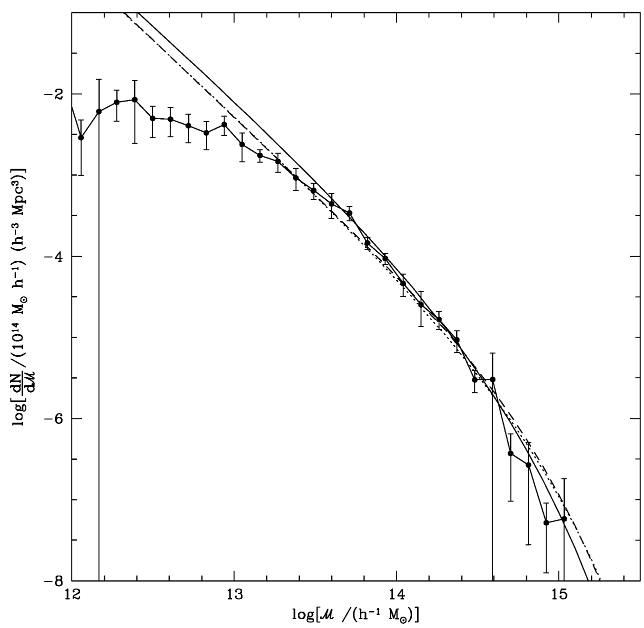

Figure 7. Group mass function for the 2dFGGC (dots). Error bars correspond to 1σ dispersion obtained from 10 mock catalogues. The solid line, dotted line and dashed line are the prescriptions of Press & Schechter (1974), Sheth & Tormen (1999) and Jenkins et al. (2001), respectively.

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6 CONCLUSIONS

Here, we have applied several statistical analyses to groups and galaxies in groups taken from an updated version of the Merchán & Zandivarez (2002) group catalogue (2dFGGC). We have focused on an accurate determination of the luminosity functions of galaxies in groups, and the luminosity and mass functions of groups, taking advantage of the statistical power of the 2dFGGC.

In the LF computations, we have used a version of the Choloniewski (1987) approach to the C^- method by Lynden-Bell (1971), adapted to the particular sky coverage of the 2dFGRS. The choice of this particular estimator of the luminosity function is inspired in the conclusions of the comparative analysis of LF estimators by Willmer (1997), who states that the C^- method is less affected by inhomogeneities in the sample. The resulting luminosity function for galaxies in groups (Fig. 1) is well fitted by a Schechter function with shape parameters $M^* - 5\log(h) = -19.90 \pm 0.03$ and $\alpha = -1.13 \pm 0.02$ as determined by a STY fitting procedure. These values are quite consistent with those obtained by Norberg et al. (2001) for field galaxies in the 2dFGRS.

We have performed a similar analysis in subsamples of galaxies (Fig. 2 and Table 1) defined by the spectral types of Madgwick et al. (2002). In general, the characteristic magnitudes M^* are shifted to higher luminosities with respect to the field values found by Madgwick et al. (2002), irrespective of the spectral type. This shift may be caused by galaxy dynamical interactions such as mergers, which are expected to be much more frequent in systems with low-velocity dispersions as the majority of our group sample.

The faint end slopes of the luminosity functions, α, are consistent with those corresponding to field galaxies except for low star-forming type 1 galaxies, which show a steeper value in groups (Fig. 3). We deepen our analysis for type 1 galaxies exploring the behaviour of α for two subsets of groups: low ($\mathcal{M} \leqslant 10^{14}\,h^{-1}\,\mathrm{M}_\odot$) and high ($\mathcal{M} > 10^{14}\,h^{-1}\,\mathrm{M}_\odot$), virial masses. We observe an increase in the faint end slope of the type 1 LF ($\alpha = -1.10 \pm 0.06$) for galaxies in high-mass systems, meanwhile for galaxies in low-mass systems it remains closer to the global value ($\alpha = -0.71 \pm 0.04$) (Fig. 4). This effect could be the result of internal processes in higher-mass system environments such as ram-pressure and galaxy harassment, which are not expected to be significantly important in smaller overdensities.

We have defined group luminosities following Moore et al. (1993), as the sum of its observed luminosity plus a normalized integral of the galaxy luminosity function below the flux limit of the survey. This definition has proved to be reliable in the computation of group total luminosity using observations and artificial flux and volume-limited catalogues. The main aim of this computation is the determination of the luminosity function for our group catalogue. Since the C^- method for luminosity function estimation requires a fair selection criterion, we have used the Rauzy (2001) T_C statistics to determine an apparent magnitude cut-off for the 2dFGGC that ensures the highest level of completeness. Our final flux-limited group sample comprises 922 groups with apparent magnitudes brighter than $b_J = 15.6$ (Fig. 5). The resulting group luminosity function for this sample (Fig. 6) is consistent with Moore et al. (1993) determination for groups in the CfA redshift survey and with the semi-analytical model prediction of a Λ cold dark matter cosmology performed by Benson et al. (2000). This agreement is acceptable in the luminosity range $M_{b_J} \lesssim -21.5$. The differences observed at fainter luminosities are mainly caused by the lack of groups with fewer than four members in the 2dFGGC. The intrinsic characteristics of the group-finding algorithm used in the construction of the 2dFGGC by Merchán & Zandivarez (2002), determine that below this limit any resulting sample has an unacceptable level of contamination by spurious detections.

The flux-limited group sample adopted in the luminosity function computation is also fair enough for the determination of the group mass function. This determination was achieved by using an adapted version of the $1/V_\mathrm{max}$ that considers the sky coverage of the group sample. Finally, a comparison with analytical predictions of halo abundances is made using a simple scheme to relate group virial masses and that corresponding to the adequate overdensity in our cosmological model. The results are displayed in Fig. 7, where it can be seen a very good agreement with the mass functions of Press & Schechter (1974), Sheth & Tormen (1999) and Jenkins et al. (2001) for masses $\mathcal{M} \gtrsim 10^{13}\,h^{-1}\,\mathrm{M}_\odot$. Again, the disagreement for low masses can be attributed to absence of poor groups.

The statistical significance of the group catalogue used in this work allowed us to obtain very important clues concerning internal properties of intermediate-mass systems and also to observe the level of agreement obtained from both analytical and semi-analytical predictions within the framework of a Λ cold dark matter model.

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REFERENCES

Abadi M.G., Moore B., Bower R., 1999, MNRAS, 308, 947
Bahcall N.A., Cen R., 1993, ApJ, 407, L49
Benson A.J., Cole S., Frenk C.S., Baugh C.M., Lacey C.G., 2000, MNRAS, 311, 793
Biviano A., Girardi M., Giuricin G., Mardirossian F., Mezzetti M., 1993, ApJ, 411, L13
Blanton M.R. et al., 2001, AJ, 121, 2358
Colless M. et al. (the 2dFGRS Team), 2001, MNRAS, 328, 1039
Choloniewski J., 1987, MNRAS, 226, 273
Domínguez M.J., Zandivarez A.A., Martínez H.J., Merchán M.E., Muriel H., Lambas D.G., 2002, MNRAS, 335, 825
Eke V.R., Cole S., Frenk C.S., 1996, MNRAS, 282, 263
Garcia A.M., 1993, A&AS, 100, 47
Girardi M., Giuricin G., 2000, ApJ, 540, 45
Girardi M., Borgani S., Giuricin G., Mardirossian F., Mezzetti M., 1998, ApJ, 506, 45
Goto T. et al., 2002, PASJ, 54, 515
Huchra J.P., Geller M.J., 1982, ApJ, 257, 423
Jenkins A., Frenk C.S., White S.D.M., Colberg J.M., Cole S., Evrard A.E., Yoshida N., 2001, MNRAS, 321, 372
Lin H., Kishner R.P., Shectman S.A., Landy S.D., Oemler A., Tucker D.L., Schechter P.L., 1996, ApJ, 464, 60
Lewis I. et al. (the 2dFGRS Team), 2002, MNRAS, 334, 673

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López-Cruz O., Yee H.K.C., Brown J.P., Jones C., Forman W., 1997, ApJ, 475, 97
Loveday J., Peterson B.A., Efstathiou G., Maddox S.J., 1992, ApJ, 390, 338
Lynden-Bell D., 1971, MNRAS, 155, 95
Madgwick D. et al. (the 2dFGRS Team), 2002, MNRAS, 333, 133
Marinoni C., Hudson M., Giuricin G., 2002, ApJ, 569, 91
Martínez H.J., Zandivarez A., Dominguez M., Merchán M.E., Lambas D.G., 2002, MNRAS, 333, 31L
Marzke R.O., Huchra J.P., Geller M.J., 1994, ApJ, 428, 43
Merchán M.E., Zandivarez A., 2002, MNRAS, 335, 216
Moore B., Frenk C.S., White S.D.M., 1993, MNRAS, 261, 827
Moore B., Katz N., Lake G., Dressler A., Oemler A., 1996, Nat, 379, 613
Muriel H., Valotto C., Lambas D.G., 1998, ApJ, 506, 540
Norberg P. et al. (the 2dFGRS Team), 2001, MNRAS, 328, 64
Press W., Schechter P., 1974, ApJ, 187, 425
Ratcliffe A., Shanks T., Parker Q.A., Fong R., 1998, MNRAS, 294, 147
Rauzy S., 2001, MNRAS, 324, 51
Sandage A., Tammann G.A., Yahil A., 1979, ApJ, 352, 82
Schmidt M., 1968, ApJ, 151, 393
Sheth R.K., Tormen G., 1999, MNRAS, 308, 119
Trentham N., 1997, MNRAS, 290, 334
Valotto C.A., Nicotra M.A., Muriel H., Lambas D.G., 1997, ApJ, 479, 90
Willmer C.N.A., 1997, AJ, 114, 898
York D.G. et al., 2000, AJ, 120, 1579
Zucca E. et al., 1997, A&A, 326, 477

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